★ Box-Cox Power Transformer:

i) Box-Cox Transform:

$$x_i^{\lambda} = \begin{cases} \dfrac{x_i^{\lambda} - 1}{\lambda} & \text{if } \lambda \neq 0, \\ \ln(x_i) & \text{if } \lambda = 0 \end{cases}$$

The exponent here is a variable called lambda(λ) that varies over the range of -5 to 5, and in the process of searching, we examine all values of λ. Finally, we choose the optimal value (resulting in the best approximation to normal distribution) for your variable.

Applyed only for ~~$x \geq 0$~~ $x > 0$

Techniques for this: • Max likelihood
• Baysian

ii) Yeo-Johnson Transform

$$x_i^{(\lambda)} = \begin{cases} [(x_i+1)^{\lambda} - 1]/\lambda & \text{if } \lambda \neq 0, \; x_i \geq 0; \\ \ln(x_i) + 1 & \text{if } \lambda = 0, \; x_i \geq 0; \\ -[(-x_i+1)^{2-\lambda} - 1]/(2-\lambda) & \text{if } \lambda \neq 2, \; x_i < 0; \\ -\ln(-x_i+1) & \text{if } \lambda = 2, \; x_i < 0; \end{cases}$$

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This transformation is some what of an adjustment to Box-cox transformation, by which we can apply it to negitive numbers.

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